

LSTM based Method for Forecasting Hyperlocal Air Quality in Metropolitan Cities

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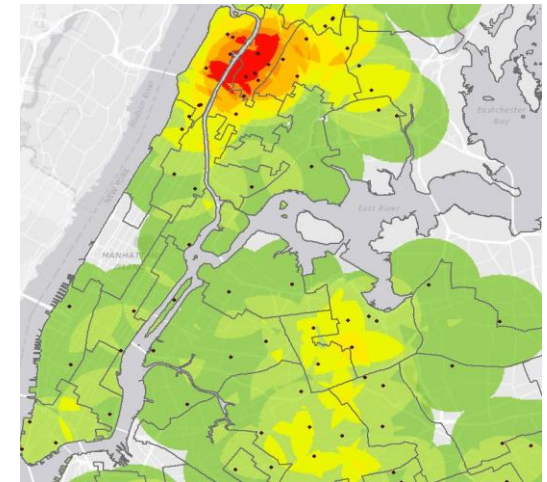
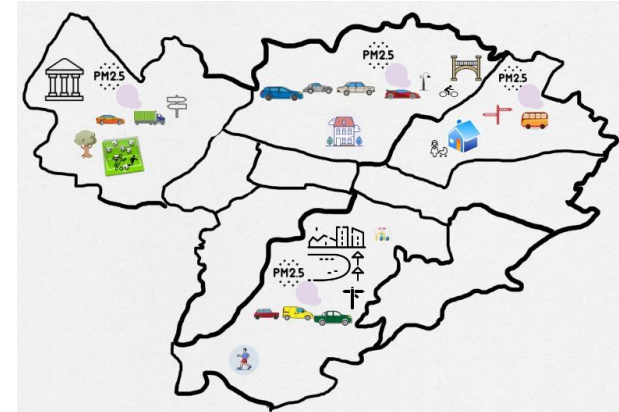
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Outline

- Introduction
 - Motivating scenario
 - Hyperlocal data challenges
- Methodology
- Results & Evaluation
 - Model Evaluation
- Conclusion
 - Concluding remarks

Motivating scenario

- Billions of GPS-enabled handheld devices collect massive amounts of data
 - Urban planning, Transportation data, Air Quality data, Meteorological etc.
- Analyze data to guide decision making
 - Smart City planning (e.g. Infrastructure, facilities, walkability)
- **Geospatial analysis**
 - Air pollution **density** in each **zone**
 - Autocorrelation between spatial proximity, time and pollution

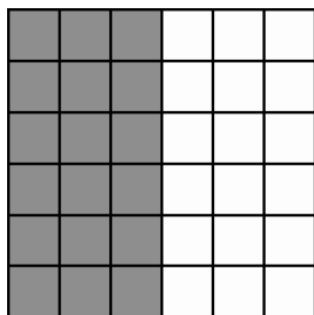


Outline

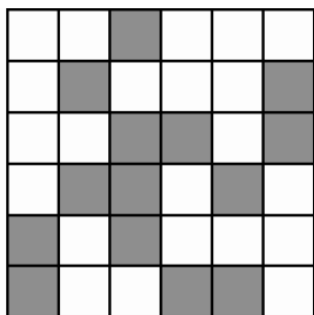
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Tobler's First Law of Geography

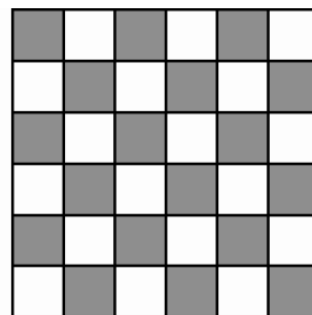
- “**Everything** is **related** to everything else, but **near things** are **more related** than distant things.”
 - **Autocorrelation Analysis**



Positive spatial
autocorrelation



No spatial
autocorrelation

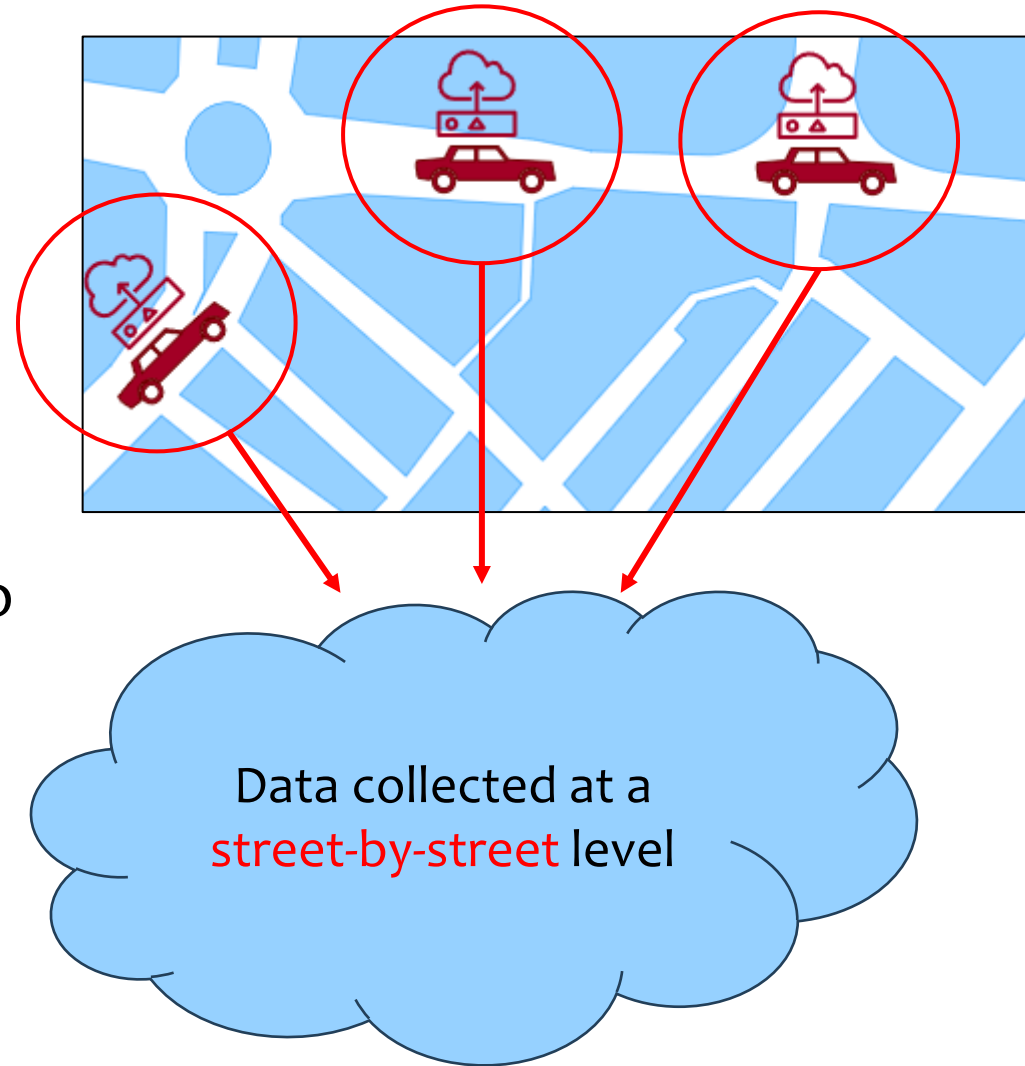


Negative spatial
autocorrelation

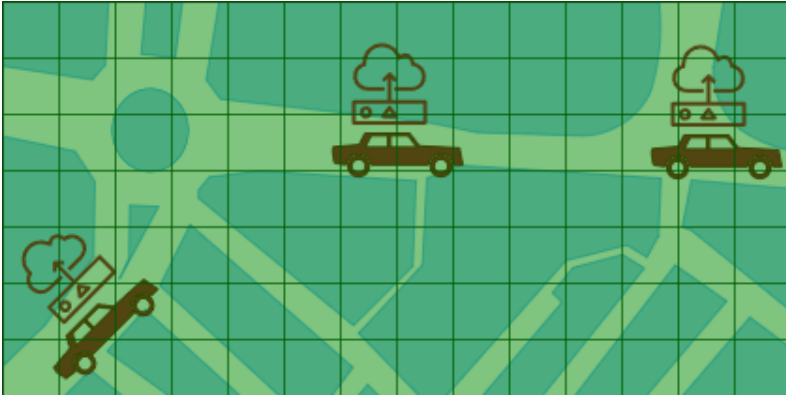
[Image source](#)

Hyperlocal data challenges

- Optimal **spatial representation** is often **context dependent**
 - Empirically determined
- Optimal **forecasting model** to use depends on the **data's nature**
 - Requires testing and evaluation



Optimal Geohash encoding



Is this representation optimal?



Or this representation?

Higher Resolution is not always the best option

Geohash can be used to group data points

- Each geohash is a **strata**
- All geohash covering the area are **stratum**

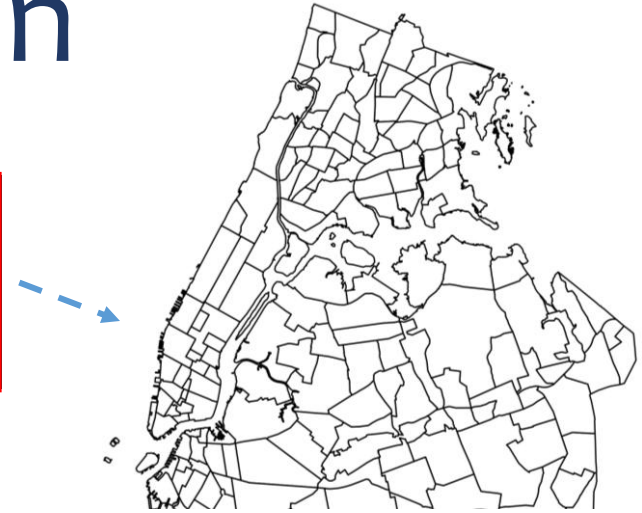
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Geospatial Join

GeoJSON file

neighborhood	boroughCode	borough	geometry
Allerton	2	Bronx	POLYGON ((-73.8486 40.87167, -73.84582 40.8702...
Alley Pond Park	4	Queens	POLYGON ((-73.74333 40.73888, -73.74371 40.739...
Arden Heights	5	Staten Island	POLYGON ((-74.16983 40.56108, -74.16982 40.561...



NYC AQ csv data

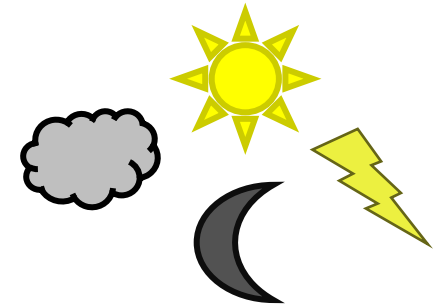
time	latitude	longitude	temperature	humidity	pm25
1631277304	40.847672	-73.869316	23.7	57.3	4.508813
1631277308	40.847668	-73.869316	23.7	57.8	5.462420
1631277313	40.847649	-73.869362	23.7	57.8	5.154881

Polygons

Points

(parametrized)

Projected Coordinate System (PCS)



time	latitude	longitude	temperature	humidity	pm25	geometry	neighborhood	borough
1631278851	40.845364	-73.870583	21.8	64.0	5.099380	POINT (-73.87058 40.84536)	Van Nest	Bronx
1631278856	40.845364	-73.870583	21.8	63.8	5.216026	POINT (-73.87058 40.84536)	Van Nest	Bronx
1632320946	40.745548	-73.844963	29.6	60.2	12.031301	POINT (-73.84496 40.74555)	Flushing Meadows Corona Park	Queens

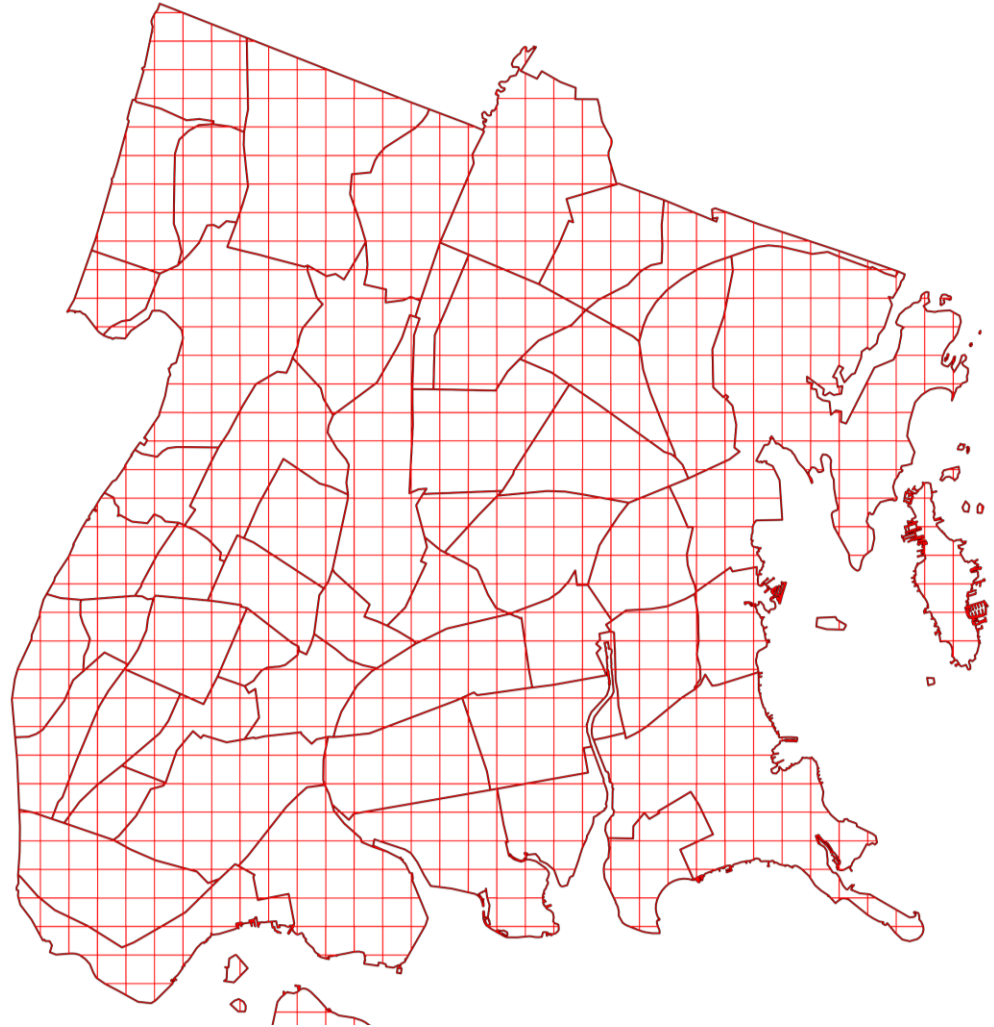
Resulting Output of Geospatial Join

Spatial Indexing (Geohash)

Geohash is a **Hierarchical spatial indexing** method that encodes the space recursively.

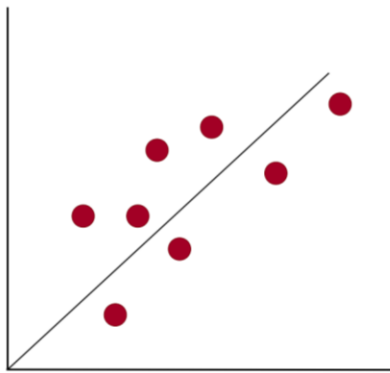
It facilitates quick lookup, progressive zooming and **spatial grouping**

Spatial resolution is controlled by the **precision** parameter.

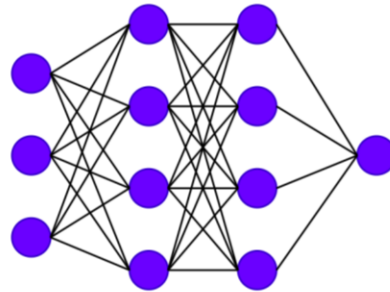


Model Selection

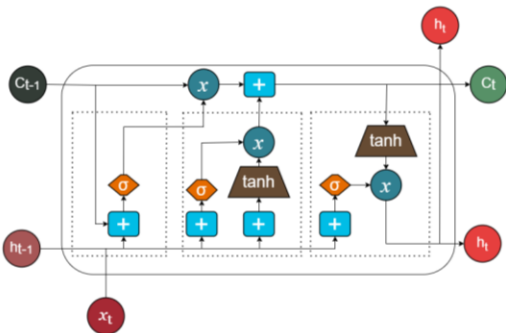
- Several models were considered for forecasting **PM_{2.5}**
 - Linear Regression, Support Vector Regression (SVR)
 - Multi-Layer Perceptron (MLP), Recurrent Neural Network (RNN) and **Long-Short Term Memory (LSTM)**



❌ **Linear Regression**
Fails to capture non-linear patterns in data



❌ **MLP**
Treats all inputs independently

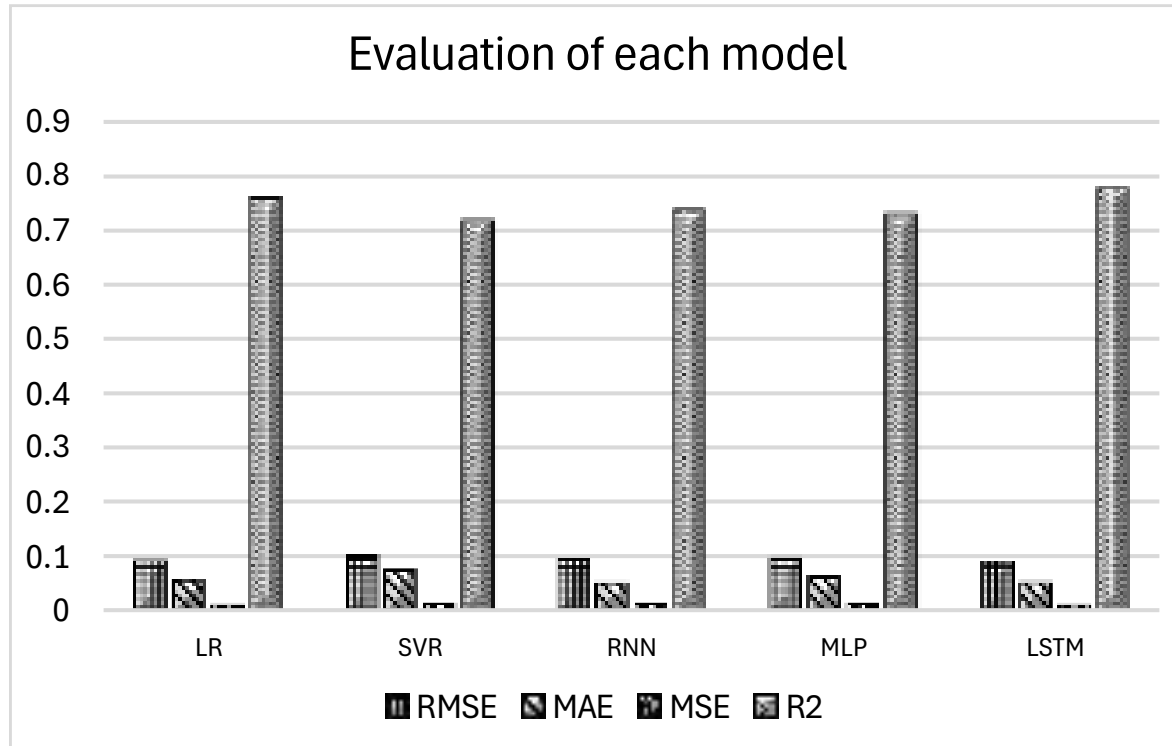


★ **LSTM**
Good choice for time-series data

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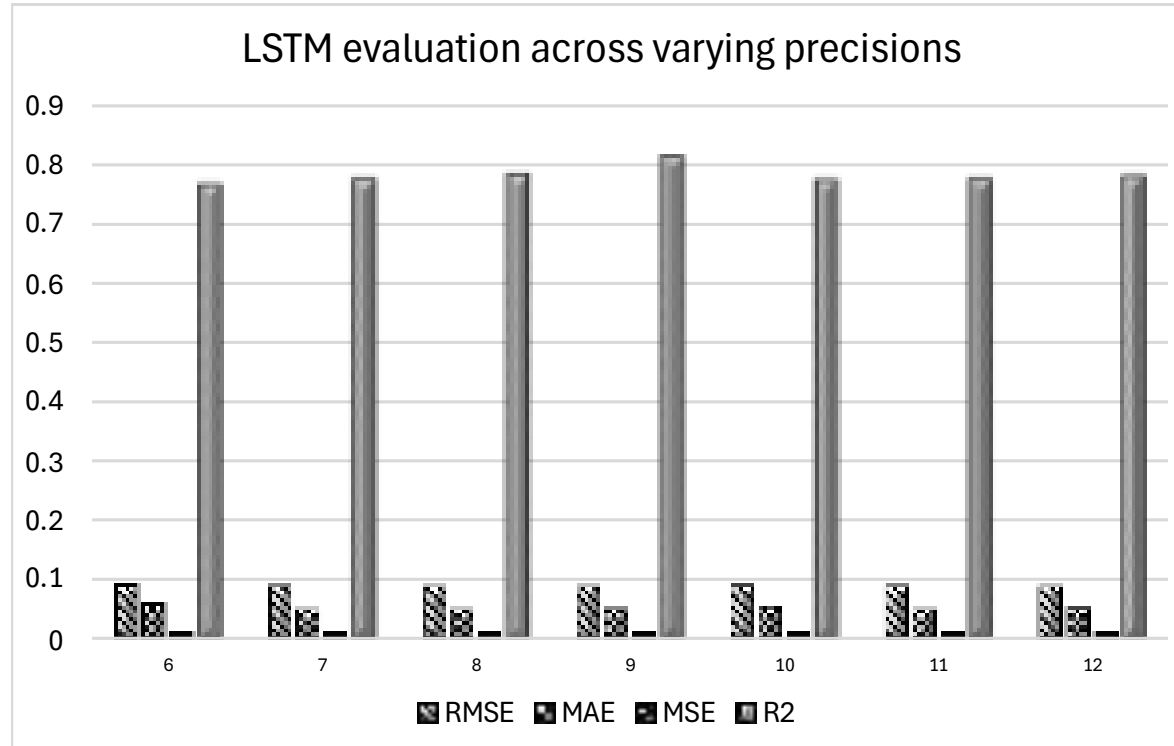
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Model evaluation on NYC AQ data



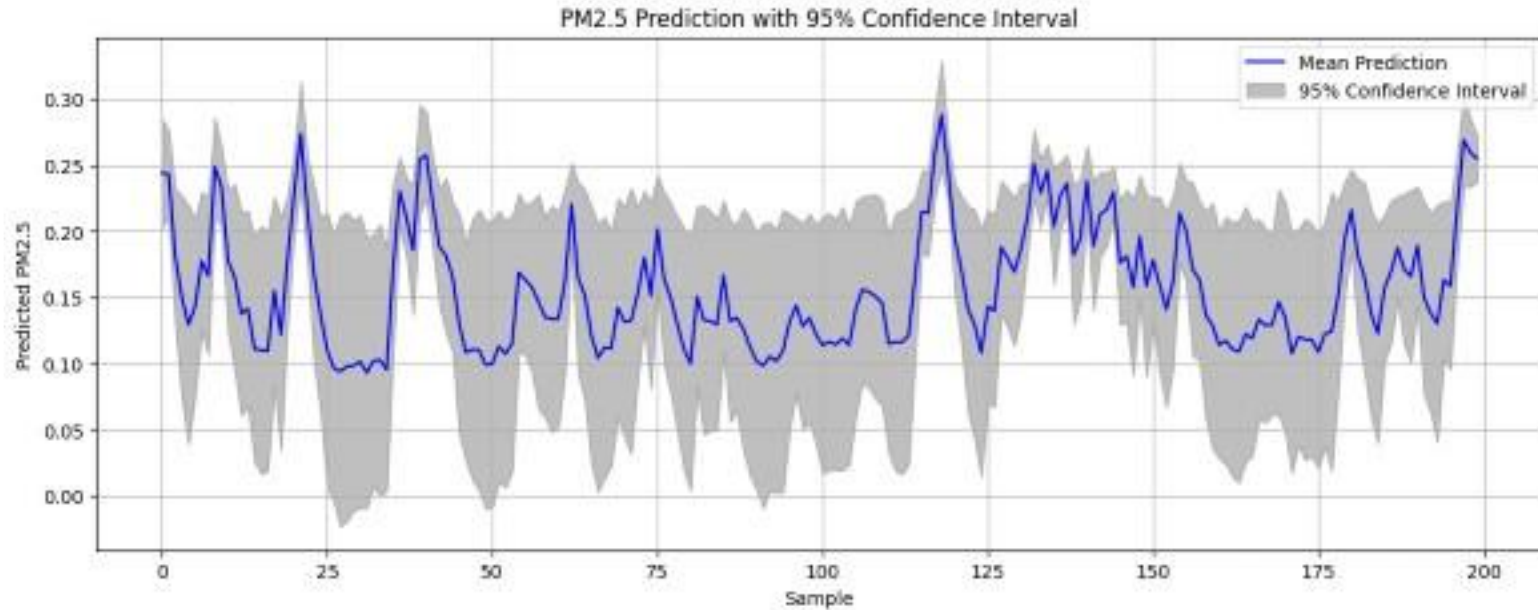
- Measuring the performance of **several** models
- Without geospatial join and without geospatial encoding
- Results are subpar, with LSTM outperforming other models

Model evaluation of LSTM



- Measuring the performance of **LSTM network**
- With **geospatial join** and **varying** the geohash precision
- Geohash precision 9, achieves best results
- Improved R2 and lower error rates

Confidence of LSTM predictions



- Evaluating confidence of **LSTM network predictions**
- Use Monte Carlo dropout at inference time
- Confidence interval generated withing 50 stochastic passes
- Measure of model reliability

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Conclusion

- **This end-to-end pipeline** is a system designed for smart city scenarios **which is applied to work with Geo-referenced hyperlocal data of varying resolutions.**
 - Spatial analysis is conducted via a **Geospatial Join** operation.
 - Data grouping using **Spatial Indexing (Geohash)** with varying resolution by tuning the **precision** parameter.
 - LSTM used for forecasting PM_{2.5} due to its ability to capture long term dependencies.
 - Evaluation using statistical metrics and Monte Carlo dropout.

Q&A

Thanks for your attention!
Question's time...

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